

PASINDU INDUWARA

Software Engineer

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PROFILE

Software Engineer at Marketrix Inc., building AI agents and QA automation products end to end, from architecture through to production. Previously a Software Engineer Intern at WSO2, and a continuing contributor to OpenChoreo, an open source Internal Developer Platform. My work has moved from platform infrastructure into agentic systems, and my final-year research extends that line into autonomous incident response for cloud-native environments.

WORK EXPERIENCE

SOFTWARE ENGINEER | MARKETRIX INC., SAN FRANCISCO (REMOTE) | JANUARY 2026 – PRESENT

Marketrix builds an AI user-research and QA platform that runs web products through simulated users. I own features end to end, from architectural alignment through implementation and testing to production rollout.

- **Persona simulation:** Study engines for survey, user-research, and A/B runs, covering automated page parsing, persona-driven form filling, and aggregation of responses into reportable themes.
- **Analytics and reporting:** Conversion pipeline reporting lift with 95% confidence intervals and projections onto partner traffic volumes, and a vector charting layer that replaced screenshot capture in reports.
- **Autonomous QA:** Self-improving test loop that generates, runs, judges, and rewrites tests until they verify, repairing them against retrieved product knowledge, with each stage traced and scored offline.
- **Workflow automation:** Recurring and event-driven automation with cron scheduling, a standalone worker, and run history, triggered from team chat commands and pull request events, with multi-channel alerting.
- **Platform and security:** Tenant isolation across customer data access, credentials encrypted and write-only at the service boundary, and full-table scans replaced with targeted partial indexes.

SOFTWARE ENGINEER INTERN | WSO2 | JULY 2025 – DECEMBER 2025

- Designed the architecture for the Choreo Backstage plugin and scaffolded the full-stack project structure, after deploying a connected application on Choreo and integrating the runner plugin into Backstage.
- Built the Choreo and OpenChoreo Incremental Entity Provider, mapping platform resources onto catalogue entities with mark-and-sweep cleanup and delta updates that apply only changed resources per sync.
- Authored the design proposal for and implemented cursor pagination across the OpenChoreo APIs, returning bounded pages with opaque continuation tokens and explicit expiry handling.
- Extended the choreoctl command line tool and catalogue clients to paginate automatically, with integration testing against the new response structures and the supporting documentation.

RESEARCH

FINAL-YEAR RESEARCH PROJECT | UNIVERSITY OF KELANIYA | 2026 – PRESENT

“Towards Autonomous Cloud Security: Evaluating a Hierarchical LLM-Based Multi-Agent Framework for Incident Response in Cloud-Native Environments” – designing and evaluating a hierarchical framework in which LLM-based agents coordinate across the incident response lifecycle, from detection and triage through to remediation.

OPEN SOURCE CONTRIBUTIONS

- **OpenChoreo** – Contributor to the open source Internal Developer Platform. Designed and implemented cursor-based pagination across the public interface, internal clients, and the choreoctl command line tool, replacing a return-everything strategy with resumable continuation tokens, explicit handling of expired and invalid cursors, transparent automatic paging for callers, and a result-size cap. Also contributed the changes backing the Backstage incremental entity module.

- **SST (Serverless Stack)** – Integrated additional AI providers and model configurations into the models.dev catalogue.
- **KiloCode** – Documentation on model-initiated slash commands and editor interactions.

EDUCATION

UNIVERSITY OF KELANIYA | 2023 – PRESENT | EXPECTED GRADUATION – OCTOBER 2026
B.Sc. Honours in Information Technology, Current GPA: 3.7

PROFESSIONAL QUALIFICATIONS

TRAINEE FULL STACK DEVELOPER | University of Moratuwa | Centre for Open & Distance Learning
Completed full-stack developer training covering Python programming, web design, and server-side development.

PROJECTS

1. **Guidia – Full-Stack Career Guidance Platform (Individual)** – Platform digitising the University of Kelaniya’s recruitment and counselling processes. React, TypeScript, and Zod frontend over an Express and MySQL backend with Socket.IO messaging, on Azure Container Apps and Blob Storage.
2. **Browtrix – MCP Browser Automation Ecosystem (Individual)** – Real-time ecosystem bridging AI assistants and web browsers over the Model Context Protocol. Python FastMCP, FastAPI, and WebSockets backend in a Turbo monorepo; Next.js 15 and React 19 frontend with HTML snapshotting, confirmation modals, and connection pooling for human-in-the-loop workflows.
3. **Quota.app – Fuel Quota Management System (Group)** – Microservices fuel system for vehicle owners, stations, and administrators. Spring Boot backend with JWT and JPA, three Next.js frontends and an Expo/React Native app, QR transaction processing, Twilio SMS, and PostgreSQL reporting.

ACHIEVEMENTS

- **First Runner-Up** – J’PURA EXPO 2023, Inter University Export-Oriented Innovation Competition.
- **Semi-Finalist** – VentureVerse Startup Challenge, Sabaragamuwa University, for the project “Ceylon Treasure”.

VOLUNTEER WORK AND AFFILIATIONS

- Assistant Media Director, Industrial Management Science Students’ Association (IMSSA), 2024 – 2025
- Member, AIESEC Colombo North Local Committee, 2023 – 2025
- Member, Gavel Club – University of Kelaniya, 2023 – 2025
- Volunteer, Sasnaka Sansada Foundation, 2022 – 2024

SKILLS

- **Languages:** TypeScript, JavaScript, Python, Go, Java, SQL, HTML, CSS
- **Backend & Data:** Node.js, Express, gRPC, FastAPI, Spring Boot, Sequelize, BullMQ, PostgreSQL, MySQL, MongoDB, Redis
- **Frontend:** React, Next.js, Tailwind CSS, Zod, Socket.IO, Recharts, React Native, Expo
- **AI & Evaluation:** LiteLLM, LangChain, RAG, LLM-as-a-judge evaluation, MLflow, Model Context Protocol (MCP), FAISS, PyTorch
- **Infrastructure & DevOps:** Docker, Kubernetes, Terraform, Helm, Azure (Container Apps, Blob Storage, DevOps), Firebase, Supabase, Git, GitHub Actions
- **Testing & QA:** Jest, Pytest, Vitest, Selenium, TestNG, Postman, cross-browser testing, CI/CD test integration

LANGUAGES

- English (Fluent) Sinhala (Native Speaker)

REFERENCES

Available on request.